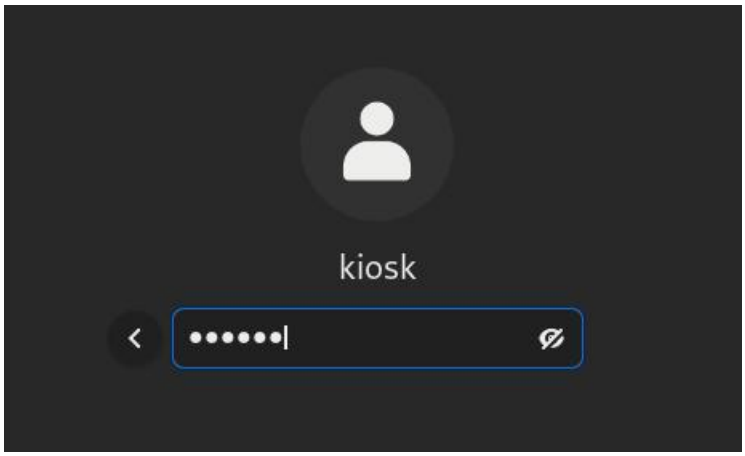
RHEL9 课堂练习环境部署方法-bgl

1. 笔记本要求

2. 安装 VMware17(解压后会看到序列号文件)
3. 打开虚拟机 RHEL9(移动而来)



①物理机 foundation0 上的账户及密码（练习环境以 root 或 kiosk 的身份登录均可）

root Asimov

kiosk redhat

student student

②子 Linux servera 或 serverb 上的账户及密码

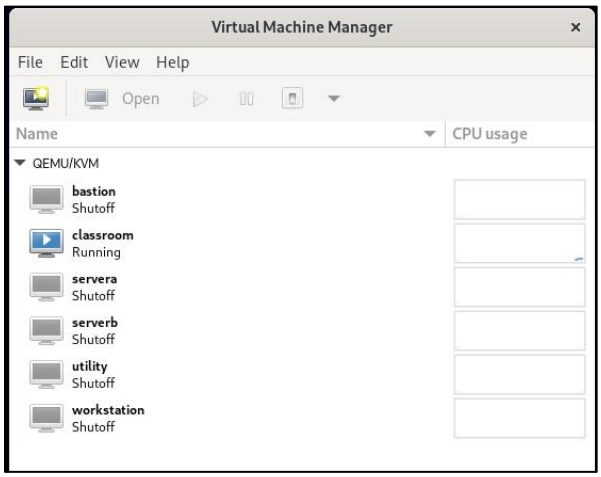
root redhat

4. RHCE9 环境架构及控制各台子 Linux(servera,serverb 等)的方法

RHCE9 环境架构：

物理机 foundation0--->Virtual Machine Manager--->bastion(网关服务器)
workstation(工作站服务器)
classroom(环境服务器)
utility(容器镜像服务器)
servera(练习机 a)

serverb(练习机 b)



控制各台子 Linux(servera,serverb 等)的方法：

在物理机 foundation0 上运行的管理命令：

rht-vmctl status all　　// 一定要保证 bastion，classroom 处于运行状态

rht-vmctl status classroom　　//虚拟机 classroom 需要单独查看

rht-vmctl start all　　//启动 servera 和 serverb 需要 5 分钟，耐心等待

rht-vmctl reset servera　　//重置虚拟机 servera

rht-vmctl fullreset all -q　　//全部重置虚拟机，无需交互

rht-vmctl stop all　　//停止各个虚拟机需要 2 分钟时间

启动各台虚拟机后，可以在 foundation 上远程登录 servera 和 serverb

```
[kiosk@foundation0 ~]$ rht-vmctl status all
bastion DEFINED
workstation DEFINED
utility DEFINED
servera DEFINED
serverb DEFINED
[kiosk@foundation0 ~]$ rht-vmctl status classroom
classroom RUNNING
[kiosk@foundation0 ~]$ rht-vmctl start all
Starting bastion.
Starting workstation.
Starting utility.
Starting servera.
Starting serverb.
[kiosk@foundation0 ~]$ rht-vmctl status all
bastion RUNNING
workstation RUNNING
utility RUNNING
servera RUNNING
serverb RUNNING
[kiosk@foundation0 ~]$ ping servera
PING servera.lab.example.com (172.25.250.10) 56(84) bytes of data.
From foundation0.ilt.example.com (172.25.250.250) icmp_seq=1 Destination Host Unreachable
From foundation0.ilt.example.com (172.25.250.250) icmp_seq=2 Destination Host Unreachable
From foundation0.ilt.example.com (172.25.250.250) icmp_seq=3 Destination Host Unreachable
^C
```

启动各台虚拟机需要几分钟，耐心等待

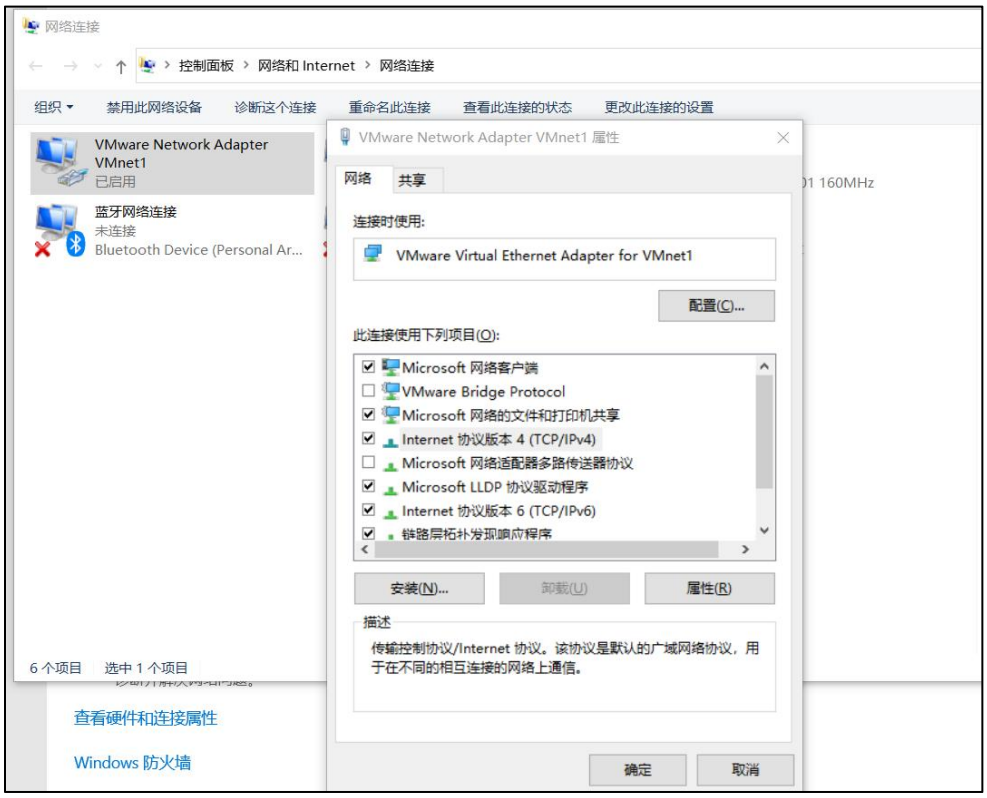
```
[kiosk@foundation0 ~]$ ping servera
PING servera.lab.example.com (172.25.250.10) 56(84) bytes of data.
From foundation0.ilt.example.com (172.25.250.250) icmp_seq=1 Destination Host Unreachable
From foundation0.ilt.example.com (172.25.250.250) icmp_seq=2 Destination Host Unreachable
From foundation0.ilt.example.com (172.25.250.250) icmp_seq=3 Destination Host Unreachable
From foundation0.ilt.example.com (172.25.250.250) icmp_seq=4 Destination Host Unreachable
From foundation0.ilt.example.com (172.25.250.250) icmp_seq=5 Destination Host Unreachable
From foundation0.ilt.example.com (172.25.250.250) icmp_seq=6 Destination Host Unreachable
From foundation0.ilt.example.com (172.25.250.250) icmp_seq=7 Destination Host Unreachable
From foundation0.ilt.example.com (172.25.250.250) icmp_seq=8 Destination Host Unreachable
From foundation0.ilt.example.com (172.25.250.250) icmp_seq=9 Destination Host Unreachable
64 bytes from servera.lab.example.com (172.25.250.10): icmp_seq=10 ttl=64 time=1377 ms
64 bytes from servera.lab.example.com (172.25.250.10): icmp_seq=12 ttl=64 time=0.239 ms
64 bytes from servera.lab.example.com (172.25.250.10): icmp_seq=13 ttl=64 time=0.674 ms
64 bytes from servera.lab.example.com (172.25.250.10): icmp_seq=14 ttl=64 time=0.596 ms
^C
```

ssh root@servera，密码 redhat

5. 如何在 Windows 系统 ping 通 foundation0

```
[kiosk@foundation0 ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq master br0 state UP group default qlen 1000
    link/ether 00:0c:29:82:40:ae brd ff:ff:ff:ff:ff:ff
    altname enp3s0
3: br1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 52:54:00:2a:d9:11 brd ff:ff:ff:ff:ff:ff
    inet 192.168.123.1/24 brd 192.168.123.255 scope global br1
       valid_lft forever preferred_lft forever
4: br0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 00:0c:29:82:40:ae brd ff:ff:ff:ff:ff:ff
    inet 172.25.254.250/24 brd 172.25.254.255 scope global noprefixroute br0
       valid_lft forever preferred_lft forever
    inet 172.25.0.250/24 brd 172.25.0.255 scope global noprefixroute br0
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe82:40ae/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

在 Windows 的 VMware Network Adapter VMnet1 的 TCP/IPv4 中添加一个 172.25.0 网段的 IP 地址，只设置 IP 地址（172.25.0.111）和子网掩码（255.255.255.0）。



此时，再次在 Windows 的 MS-DOS 中就可以 ping 通 foundation0（172.25.0.250）了。

```
C:\Users\hp>ping 172.25.0.250

正在 Ping 172.25.0.250 具有 32 字节的数据:
来自 172.25.0.250 的回复: 字节=32 时间<1ms TTL=64
来自 172.25.0.250 的回复: 字节=32 时间<1ms TTL=64
来自 172.25.0.250 的回复: 字节=32 时间<1ms TTL=64
来自 172.25.0.250 的回复: 字节=32 时间<1ms TTL=64

172.25.0.250 的 Ping 统计信息:
    数据包: 已发送 = 4，已接收 = 4，丢失 = 0 (0% 丢失)，
往返行程的估计时间(以毫秒为单位):
    最短 = 0ms，最长 = 0ms，平均 = 0ms

C:\Users\hp>
```

6. 安装 MobaXterm 并连接到 foundation0

Windows 系统通过 MobaXterm(或 xshell)连接 foundation0，先安装 MobaXterm，然后能 ping 通的前提下，运行 ssh kiosk@172.25.0.250，密码 redhat，登录 foundation0。

注意：ssh root@172.25.0.250 在红帽 9 下是不可以的。

```
2. kiosk@foundation0:~

14/02/2024  11:31.56  /home/mobaxterm  ping 172.25.0.250

Pinging 172.25.0.250 with 32 bytes of data:
Reply from 172.25.0.250: bytes=32 time<1ms TTL=64
Reply from 172.25.0.250: bytes=32 time=1ms TTL=64
Reply from 172.25.0.250: bytes=32 time<1ms TTL=64


14/02/2024  11:32.50  /home/mobaxterm  ssh kiosk@172.25.0.250
Warning: Permanently added '172.25.0.250' (ED25519) to the list of known hosts.
kiosk@172.25.0.250's password:
Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Wed Feb 14 10:25:33 2024
/usr/bin/xauth:  file /home/kiosk/.Xauthority does not exist
[kiosk@foundation0 ~]$
```

7. 关闭多虚拟机环境

在 foundation0 的终端窗口运行 poweroff